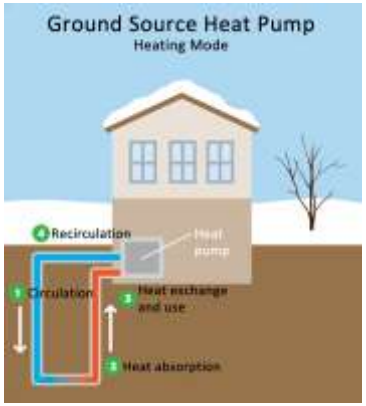
New Energies for Sustainable Development

Renewable Energy



Renewable Energy

- Maine is endowed with plentiful bioenergy, wind, hydropower, ocean, and other renewable energy resources.
- Non-hydro renewables are responsible for 32% of in-state generation, a higher percentage than in any other state in the nation.

Renewable Energy

- Tidal energy device to generate electricity for the power grid and first working offshore wind turbine.

Solar Energy

We use solar thermal energy systems to

- heat water for use in homes, buildings, or swimming pools
- heat the inside of homes, greenhouses, and other buildings
- heat fluids to high temperatures in solar thermal power plants

Solar photovoltaic devices, or solar cells, change sunlight directly into electricity.

Solar Energy

Solar Energy

The two main <u>benefits</u> of using solar energy are

- Systems do not produce air pollutants or carbon dioxide.
- Systems on buildings have minimal impact on the environment.

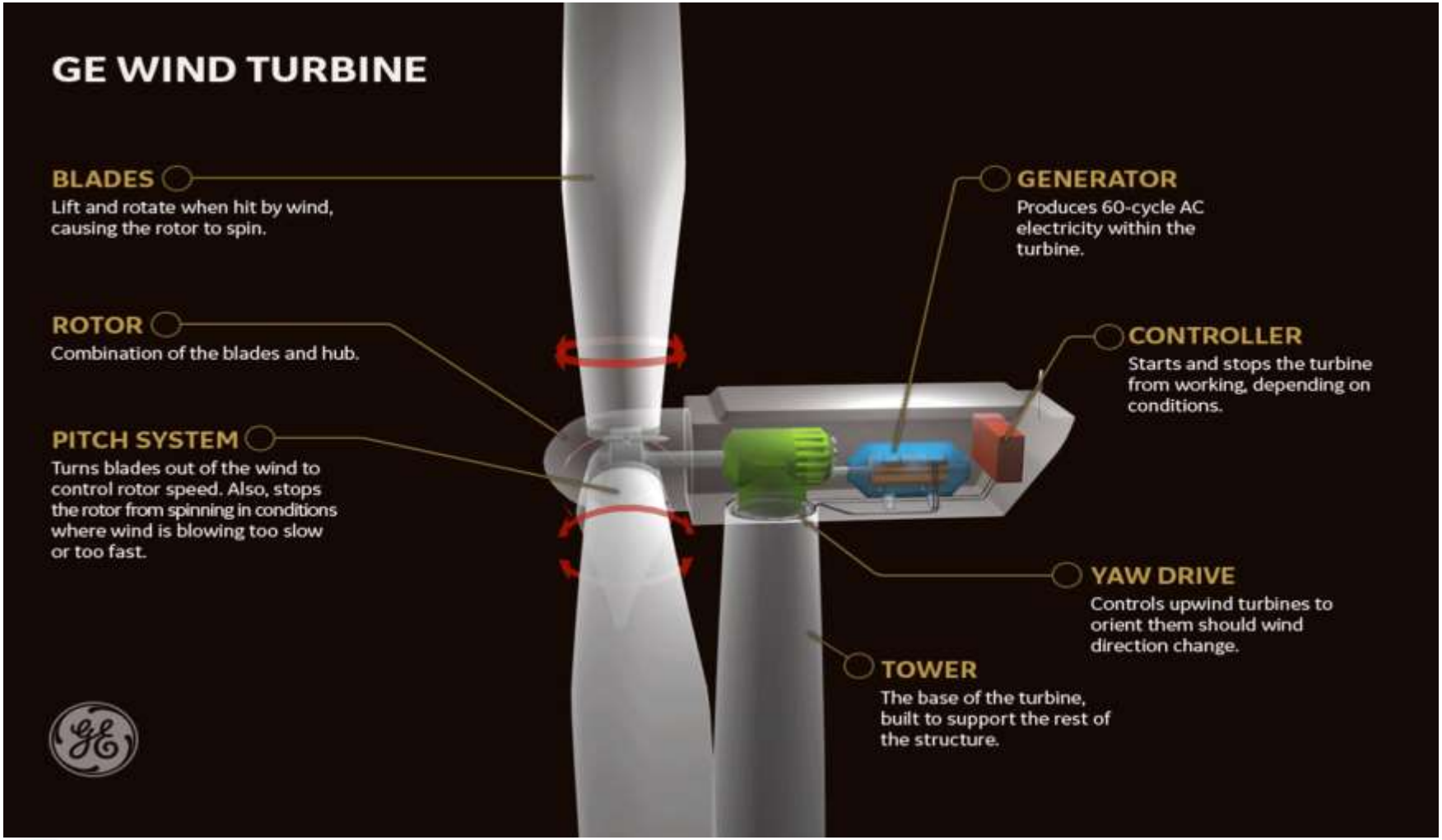
The main <u>limitations</u> of solar energy are

- The amount of sunlight that arrives at the earth's surface is not constant. The amount of sunlight varies depending on location, time of day, season of the year, and weather conditions.
- The amount of sunlight reaching a square foot of the earth's surface is relatively small, so a large surface area is necessary to absorb or collect a useful amount of energy.

Wind Energy

- Wind turbines operate on a simple principle. The energy in the wind turns two or three propeller-like blades around a rotor. The rotor is connected to the main shaft, which spins a generator to create electricity.

Wind Energy


GE WIND TURBINE
BLADES
Lift and rotate when hit by wind, causing the rotor to spin.
ROTOR
Combination of the blades and hub.
PITCH SYSTEM
Turns blades out of the wind to control rotor speed. Also, stops the rotor from spinning in conditions where wind is blowing too slow or too fast.
GENERATOR
Produces 60-cycle AC electricity within the turbine.
CONTROLLER
Starts and stops the turbine from working, depending on conditions.
YAW DRIVE
Controls upwind turbines to orient them should wind direction change.
TOWER
The base of the turbine, built to support the rest of the structure.

Wind Energy

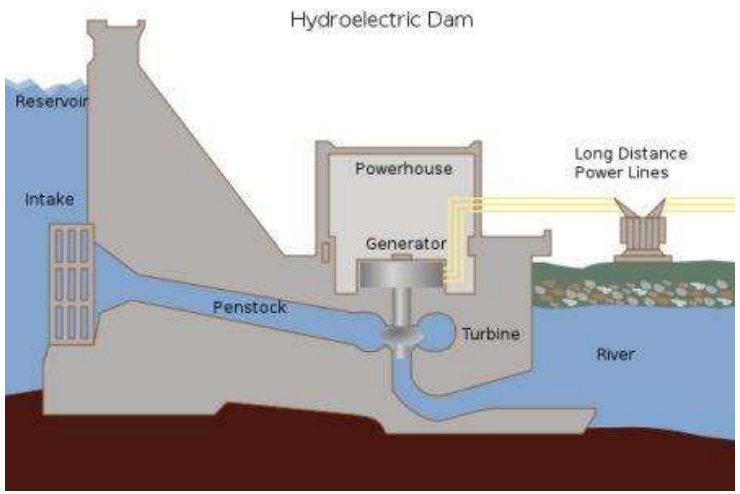
- Wind turbines are mounted on a tower to capture the most energy. At 100 feet or more above ground, they can take advantage of faster and less turbulent wind.
- Wind turbines can be used to produce electricity for a single home or building, or they can be connected to an electricity grid for more widespread electricity distribution.

Wind Energy

- Advantages of Wind Energy
  - Clean and renewable source of power
  - Cost effective
  - Rapid growth of industry, large potential
- Disadvantages of Wind Energy
  - Wind reliability
  - Threat to wildlife
  - Noise and visual pollution

Tidal Energy

- Tidal Stream Generator
  - Makes use of the kinetic energy of moving water to power turbines, in a similar way to wind turbines that use wind to power turbines.
- Tidal Barrage
  - Tidal barrages make use of the potential energy in the difference in height between high and low tides.



Tidal Energy

- Advantages
  - Clean fuel source compared to fossil fuels
  - Domestic source of energy
- Disadvantages
  - Tidal power can have effects on marine life.
    - The turbines can accidentally kill swimming sea life with the rotating blades.
    - Some fish may no longer utilize the area if threatened with a constant rotating or noise-making object.
    - Installing a barrage may change the shoreline within the bay or estuary, affecting a large ecosystem that depends on tidal flats.

Wave Energy

- Ocean waves contain tremendous energy potential.
- Wave power devices extract energy from the surface motion of ocean waves or from pressure fluctuations below the surface.

Wave Energy

- Advantages
  - Renewable
  - Environmentally friendly compared to fossil fuel energy
  - Variety of designs to use
  - Less energy dependence from foreign governments
- Disadvantages
  - Can affect the marine environment
  - May disturb private or commercial shipping
  - Dependent on wavelength for best operation
  - Poor performance in rough weather
  - Visual/noise issues

Geothermal Energy

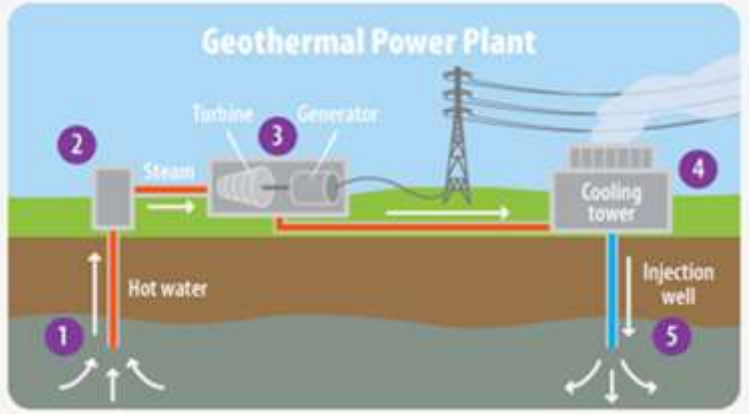
- (geo = earth and thermal = heat), geothermal energy comes from heat produced by the Earth.

Geothermal Energy

- **Direct geothermal energy** can be accessed in areas where hot springs/geothermal reservoirs are near the surface of the Earth.
- **Geothermal heat pumps** utilizes a series of underground pipes, an electric compressor and a heat exchanger to absorb and transfer heat.
- **Geothermal power plants** also harness the heat of the Earth through hot water and steam. In these plants, heat is used to generate electricity.

Geothermal Energy

- Advantages
  - Renewable energy
  - Cleaner than burning fossil fuels
- Disadvantages
  - Cost of drilling, researching proper areas
  - Requires a suitable location



Biofuels

- The two most common types of biofuels are **ethanol** and **biodiesel**.

Biofuels

- Ethanol is an alcohol.
- Ethanol is mostly used as a fuel additive to cut down a vehicle's carbon monoxide and other smog-causing emissions.

Biofuels

- Biodiesel is made by combining alcohol (usually methanol) with vegetable oil, animal fat, or recycled cooking greases.
- It can be used as an additive to reduce vehicle emissions (typically 20%) or in its pure form as a renewable alternative fuel for diesel engines.

Biofuels

- **Advantages**
  - Easy to source
  - Renewable
  - Reduces greenhouse gases
  - Reduced dependence on foreign energy
- **Disadvantages**
  - Higher cost of production (lower supply than gasoline)
  - Monoculture
  - Shortage of food
  - Water Use

Energy Conservation

- Energy conservation is the act of saving energy by reducing a service. In other words, to conserve energy, you need to cut back on your usage. Examples include driving your car fewer kilometers per week, turning your air conditioner thermostat up a degree or two during summer and unplugging your computer or home appliances when they are not in use. In all of these examples, you are reducing the amount of energy you use by doing without or making due with less fuel or electricity.

- **IMPORTANT ASPECTS OF ENERGY CONSERVATION:**
- Conserving usable energy, which is otherwise wasted, has a direct impact on economy, environment and long-term availability of non-renewable energy sources. Energy conservation implies reduction in energy consumption by reducing losses and wastage by employing energy efficient means of generation and utilization of energy. There are three important aspects of energy conservation:

1. Economic Aspect:

- (a) **Reduction in Cost of Product** Energy conservation ultimately leads to economic benefits as the cost of production is reduced. In some energy intensive industries like steel, aluminum, cement, fertilizer, pulp and paper etc., cost of energy forms a significant part of the total cost of the product. Energy cost (expressed as percent of total cost of the product) in the entire industrial sector in India varies from as low as 0.36 per cent to as high as 65 per cent. We must strive for good energy economy using energy efficient technologies. This will reduce the manufacturing cost and lead to production of cheaper and better quality products. This is essential to stay competent with our commercial rivals.
- (b) **New Job Opportunities** Energy conservation usually requires new investments in more efficient equipment to replace old inefficient ones, monitoring of energy consumption, training of manpower, etc. Thus, energy conservation can result in new job opportunities.

2. Environmental Aspect:

- Every type of energy generation / utilization process affects the environment to some extent either directly or indirectly. The extent of degradation of environment depends mainly on the type of primary energy source. Also during any energy conversion stage a part of energy escapes to surroundings and appears in the form of heat. Thus energy is generated and utilized at the expense of adverse environmental impacts. Adoption of energy conservation means can minimize this damage.

3. Conservation of Non-Renewable Energy Assets:

- The vast bulk of energy used in the world today comes from fossil fuels, which are non-renewable. These resources were laid down many millions of years ago and are not being made any longer. This finite, non-renewable asset is being used up very fast. The quantity of fossil fuels the world community uses up in one minute what it took the earth a millennium to create. Therefore, its prices are bound to go up relative to everything else. We must abandon wasteful practices in energy utilization and conserve this resource by all means for future generations.